

Part A:

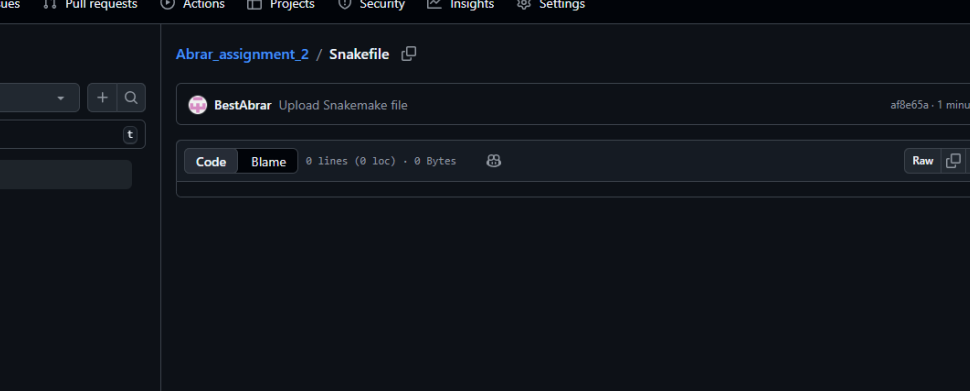
```

$ genomic_analysis_script.sh
1  #!/bin/bash
2  set -e
3
4  SRA=SRR1972739
5  REF_ID=AF086833.2
6  RESULTS_FOLDER=results
7  RAW_DIR=${RESULTS_FOLDER}/raw
8  ALIGNED_DIR=${RESULTS_FOLDER}/aligned
9  VARIANT_DIR=${RESULTS_FOLDER}/variants
10 ANNOTATED_DIR=${RESULTS_FOLDER}/annotated
11 QC_DIR=${RESULTS_FOLDER}/qc
12 SNIPEFF_DIR=${RESULTS_FOLDER}/snpeff
13 SNIPEFF_DATA_DIR=${SNIPEFF_DIR}/data/reference_db
14
15 mkdir -p $RAW_DIR $ALIGNED_DIR $VARIANT_DIR $ANNOTATED_DIR $QC_DIR $SNIPEFF_DATA_DIR
16
17 echo Downloading reference genome...
18 fetch -db nucleotide -id $REF_ID -format fasta > $RAW_DIR/reference.fasta
19 echo Downloaded reference genome!
20

```

Part B:

```
• (/home/ubuntu/BINF5506-assignment3/bioenv) ubuntu@ip-172-31-18-0:~/BINF5506-assignment3$ git commit -m "Upload Snakemake file"
[master af8e65a] Upload Snakemake file
1 file changed, 0 insertions(+), 0 deletions(-)
rename .snakemake => Snakefile (100%)
• (/home/ubuntu/BINF5506-assignment3/bioenv) ubuntu@ip-172-31-18-0:~/BINF5506-assignment3$ git push
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (2/2), 245 bytes | 245.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/BestAbrar/Abrrar_assignment_2.git
d73dfd9..af8e65a master -> master
```



Part C:

```

BINF5506-assignment3 > Snakefile
1  SRA="SRR1972739"
2  REF_ID="AF086833.2"
3  RESULTS_FOLDER="results"
4  RAW_DIR=f"{RESULTS_FOLDER}/raw"
5  ALIGNED_DIR=f"{RESULTS_FOLDER}/aligned"
6  VARIANT_DIR=f"{RESULTS_FOLDER}/variants"
7  ANNOTATED_DIR=f"{RESULTS_FOLDER}/annotated"
8  QC_DIR=f"{RESULTS_FOLDER}/qc"
9  SNPEFF_DIR=f"{RESULTS_FOLDER}/snpEff"
10 SNPEFF_DATA_DIR=f"{SNPEFF_DIR}/data/reference_db"
11 SNAKEMAKE_DIR=f"{RESULTS_FOLDER}/snakemake"
12
13 rule all:
14     input:
15         f"{SNAKEMAKE_DIR}/dirs_created"
16
17 rule create_dirs:
18     output:
19         marker = f"{SNAKEMAKE_DIR}/dirs_created"
20
21     shell:
22         """
23         echo "Makeing Pipeline files..."
24         mkdir -p {RAW_DIR} {ALIGNED_DIR} {VARIANT_DIR} {ANNOTATED_DIR} {QC_DIR} {SNPEFF_DATA_DIR}
25         touch {SNAKEMAKE_DIR}/dirs_created
26         """

```

```

(/home/ubuntu/BINF5506-assignment3/bioenv) ubuntu@ip-172-31-18-0:~/BINF5506-assignment3$ snakemake --cores 1

Select jobs to execute...

[Sun Mar 15 23:53:45 2026]
rule create_dirs:
  output: resultsSnakeMake/snakemake/dirs_created
  jobid: 1
  reason: Missing output files: resultsSnakeMake/snakemake/dirs_created
  resources: tmpdir=/tmp

Makeing Pipeline files...
[Sun Mar 15 23:53:45 2026]
Finished job 1.
1 of 2 steps (50%) done
Select jobs to execute...

[Sun Mar 15 23:53:45 2026]
localrule all:
  input: resultsSnakeMake/snakemake/dirs_created
  jobid: 0
  reason: Input files updated by another job: resultsSnakeMake/snakemake/dirs_created
  resources: tmpdir=/tmp

[Sun Mar 15 23:53:45 2026]
Finished job 0.
2 of 2 steps (100%) done
Complete log: .snakemake/log/2026-03-15T235344.678127.snakemake.log

```

```

rule download_fasta:
  input:
      f"{SNAKEMAKE_DIR}/dirs_created"

  output:
      f"{RAW_DIR}/reference.fasta"

  shell:
      """
      echo Downloading reference genome...
      efetch -db nucleotide -id {REF_ID} -format fasta > {RAW_DIR}/reference.fasta
      echo Downloaded reference genome!
      """

```

Downloading reference genome...

Downloaded reference genome!

[Mon Mar 16 00:15:33 2026]

Finished job 2.

1 of 2 steps (50%) done

Select jobs to execute...

[Mon Mar 16 00:15:33 2026]

localrule all:

input: results/snakemake/dirs_created, results/raw/reference.fasta

jobid: 0

reason: Input files updated by another job: results/raw/reference.fasta

resources: tmpdir=/tmp

[Mon Mar 16 00:15:33 2026]

Finished job 0.

2 of 2 steps (100%) done

40 echo Downloaded reference genome.

41

42 rule download_fastq:

43 input:

44 f"{SNAKEMAKE_DIR}/dirs_created"

45

46 output:

47 f"{RAW_DIR}/{SRA}.fastq"

48

49 shell:

50

51

52 echo Downloading sequencing data...

53 prefetch {SRA} -O {RAW_DIR}

54 fastq-dump -X 10000 {RAW_DIR}/{SRA}/{SRA}.sra -O {RAW_DIR}

55 echo Downloaded sequencing data!

56

Downloading sequencing data...

2026-03-16T00:22:36 prefetch.3.2.1: 1) Resolving 'SRR1972739'...

2026-03-16T00:22:36 prefetch.3.2.1: Current preference is set to retrieve SRA Normalized Format files with full base quality scores

2026-03-16T00:22:36 prefetch.3.2.1: 1) Downloading 'SRR1972739'...

2026-03-16T00:22:36 prefetch.3.2.1: SRA Normalized Format file is being retrieved

2026-03-16T00:22:36 prefetch.3.2.1: Downloading via HTTPS...

2026-03-16T00:22:38 prefetch.3.2.1: HTTPS download succeed

2026-03-16T00:22:38 prefetch.3.2.1: 'SRR1972739' is valid: 89840121 bytes were streamed from 89830148

2026-03-16T00:22:38 prefetch.3.2.1: 1) 'SRR1972739' was downloaded successfully

2026-03-16T00:22:38 prefetch.3.2.1: 1) Resolving 'SRR1972739's dependencies...

2026-03-16T00:22:38 prefetch.3.2.1: 'SRR1972739' has 0 unresolved dependencies

Read 10000 spots for results/raw/SRR1972739/SRR1972739.sra

Written 10000 spots for results/raw/SRR1972739/SRR1972739.sra

Downloaded sequencing data!

[Mon Mar 16 00:22:39 2026]

Finished job 3.

1 of 2 steps (50%) done

Select jobs to execute...

[Mon Mar 16 00:22:39 2026]

localrule all:

input: results/snakemake/dirs_created, results/raw/reference.fasta, results/raw/SRR1972739.fastq

jobid: 0

reason: Input files updated by another job: results/raw/SRR1972739.fastq

resources: tmpdir=/tmp

[Mon Mar 16 00:22:39 2026]

Finished job 0.

2 of 2 steps (100%) done

Complete log: .snakemake/log/2026-03-16T002235.703065.snakemake.log

```

rule Create_SAM:
    input:
        f"{SNAKEMAKE_DIR}/dirs_created",
        f"{RAW_DIR}/reference.fasta",
        f"{RAW_DIR}/{SRA}.fastq"

    output:
        f"{RAW_DIR}/reference.dict",
        f"{ALIGNED_DIR}/aligned.sam"

    shell:
        """
        echo Indexing reference genome with samtools...
        samtools faidx {RAW_DIR}/reference.fasta

        echo Building BWA index...
        bwa index {RAW_DIR}/reference.fasta

        echo Creating FASTA dictionary using GATK...
        gatk CreateSequenceDictionary -R {RAW_DIR}/reference.fasta -O {RAW_DIR}/reference.dict

        echo Aligning reads with read groups...
        bwa mem -R '@RG\tID:1\tLB:lib1\tPL:illumina\tPU:unit1\tSM:sample1' {RAW_DIR}/reference.fasta {RAW_DIR}/{SRA}.fastq > {ALIGNED_DIR}/aligned.sam
        echo Aligned reads!
        """

```

```

(/home/ubuntu/BINF5506-assignment3/bioenv) ubuntu@ip-172-31-18-0:~/BINF5506-assignment3$ snakemake --cores 1
[M::mem_process_seqs] Processed 10000 reads in 1.372 CPU sec, 1.372 real sec
[main] Version: 0.7.19-r1273
[main] CMD: bwa mem -R @RG\tID:1\tLB:lib1\tPL:illumina\tPU:unit1\tSM:sample1 results/raw/reference.fasta results/raw/SRR1972739.fastq
[main] Real time: 1.416 sec; CPU: 1.402 sec
Aligned reads!
[Tue Mar 17 19:08:14 2026]
Finished job 5.
1 of 2 steps (50%) done
Select jobs to execute...

[Tue Mar 17 19:08:14 2026]
localrule all:
  input: results/snakemake/dirs_created, results/raw/reference.fasta, results/raw/SRR1972739.fastq, results/qc/SRR1972739_fastqc.html, results/qc/SRR1972739_fastqc.zip, results/raw/reference.dict, results/aligned/aligned.sam
  jobid: 0
  reason: Input files updated by another job: results/raw/reference.dict, results/aligned/aligned.sam
  resources: tmpdir=/tmp

[Tue Mar 17 19:08:14 2026]
Finished job 0.
2 of 2 steps (100%) done
Complete log: .snakemake/log/2026-03-17T190809.831277.snakemake.log

```

```

rule Create_BAM:
    input:
        f"{ALIGNED_DIR}/aligned.sam"

    output:
        f"{ALIGNED_DIR}/aligned.sorted.bam",
        f"{ALIGNED_DIR}/dedup.bam",
        f"{ALIGNED_DIR}/dup_metrics.txt"

    shell:
        """
        echo Converting SAM to sorted BAM...
        samtools view -b {ALIGNED_DIR}/aligned.sam | samtools sort -o {ALIGNED_DIR}/aligned.sorted.bam

        echo Validating BAM file...
        gatk ValidateSamFile -I {ALIGNED_DIR}/aligned.sorted.bam -MODE SUMMARY

        echo Marking duplicates...
        gatk MarkDuplicates -I {ALIGNED_DIR}/aligned.sorted.bam -O {ALIGNED_DIR}/dedup.bam -M {ALIGNED_DIR}/dup_metrics.txt

        echo Indexing deduplicated BAM file...
        samtools index {ALIGNED_DIR}/dedup.bam
        """

```

```

(/home/ubuntu/BINF5506-assignment3/bioenv) ubuntu@ip-172-31-18-0:~/BINF5506-assignment3$ snakemake --cores 1
0
Indexing deduplicated BAM file...
[Tue Mar 17 19:14:27 2026]
Finished job 6.
1 of 2 steps (50%) done
Select jobs to execute...

[Tue Mar 17 19:14:27 2026]
localrule all:
  input: results/snakemake/dirs_created, results/raw/reference.fasta, results/raw/SRR1972739.fastq, results/qc/SRR1972739_fastqc.html, results/qc/SRR1972739_fastqc.zip, results/raw/reference.dict, results/aligned/aligned.sam, results/aligned/aligned.sorted.bam, results/aligned/dedup.bam, results/aligned/dup_metrics.txt
  jobid: 0
  reason: Input files updated by another job: results/aligned/dedup.bam, results/aligned/dup_metrics.txt, results/aligned/aligned.sorted.bam
  resources: tmpdir=/tmp

[Tue Mar 17 19:14:27 2026]
Finished job 0.
2 of 2 steps (100%) done
Complete log: .snakemake/log/2026-03-17T191418.916777.snakemake.log

```

```

rule Variant_Calling:
    input:
        f"{RAW_DIR}/reference.fasta",
        f"{ALIGNED_DIR}/dedup.bam"

    output:
        f"{VARIANT_DIR}/raw_variants.vcf",
        f"{VARIANT_DIR}/filtered_variants.vcf"

    shell:
        """
        echo Calling variants...
        gatk HaplotypeCaller -R {RAW_DIR}/reference.fasta -I {ALIGNED_DIR}/dedup.bam -O {VARIANT_DIR}/raw_variants.vcf
        echo Called variants!

        echo Filtering variants...
        gatk VariantFiltration -R {RAW_DIR}/reference.fasta -V {VARIANT_DIR}/raw_variants.vcf -O {VARIANT_DIR}/filtered_variants.vcf --filter
        """

```

```

(/home/ubuntu/BINF5506-assignment3/bioenv) ubuntu@ip-172-31-18-0:~/BINF5506-assignment3$ snakemake --cores 1
19:22:41.169 INFO VariantFiltration - Shutting down engine
[March 17, 2026, 7:22:41 PM UTC] org.broadinstitute.hellbender.tools.walkers.filters.VariantFiltration done. Elapsed time: 0.01 minutes.
Runtime.totalMemory()=112197632
Variants filtered!
[Tue Mar 17 19:22:41 2026]
Finished job 7.
7 of 8 steps (88%) done
Select jobs to execute...

[Tue Mar 17 19:22:41 2026]
localrule all:
  input: results/snakemake/dirs_created, results/raw/reference.fasta, results/raw/SRR1972739.fastq, results/qc/SRR1972739_fastqc.html, results/qc/SRR1972739_fastqc.zip, results/raw/reference.dict, results/aligned/aligned.sam, results/aligned/aligned.sorted.bam, results/aligned/dedup.bam, results/aligned/dup_metrics.txt, results/variants/raw_variants.vcf, results/variants/filtered_variants.vcf
  jobid: 0
  reason: Input files updated by another job: results/qc/SRR1972739_fastqc.html, results/aligned/aligned.sam, results/raw/reference.dict, results/variants/raw_variants.vcf, results/qc/SRR1972739_fastqc.zip, results/aligned/aligned.sorted.bam, results/raw/reference.fasta, results/aligned/dedup.bam, results/aligned/dup_metrics.txt, results/raw/SRR1972739.fastq, results/variants/filtered_variants.vcf, results/snakemake/dirs_created
  resources: tmpdir=/tmp

[Tue Mar 17 19:22:41 2026]
Finished job 0.
8 of 8 steps (100%) done
Complete log: .snakemake/log/2026-03-17T192154.154520.snakemake.log

```

```

rule snpEff_database:
    input:
        f"{RAW_DIR}/reference.fasta"
    output:
        f"{SNPEFF_DATA_DIR}/genes.gbk",
        f"{SNPEFF_DIR}/snpEff.config"

    shell:
        """
        echo Downloading reference GenBank file for snpEff...
        efetch -db nucleotide -id {REF_ID} -format genbank > {SNPEFF_DATA_DIR}/genes.gbk
        echo Downloaded GenBank file for snpEff!

        echo Creating custom snpEff configuration file...
        cat <<EOF > {SNPEFF_DIR}/snpEff.config
# Custom snpEff config for reference_db
reference_db.genome : reference_db
reference_db.fa : $(readlink -f {RAW_DIR}/reference.fasta)
reference_db.genbank : $(readlink -f {SNPEFF_DATA_DIR}/genes.gbk)
EOF

        echo Building snpEff database...
        snpEff build -c {SNPEFF_DIR}/snpEff.config -genbank -v -noCheckProtein reference_db
        touch {SNPEFF_DIR}/snpEff_build.done
        echo Built snpEff database!

        echo Exporting snpEff database...
        snpEff dump -c {SNPEFF_DIR}/snpEff.config reference_db > {SNPEFF_DIR}/snpEff_reference_db.txt
        echo Exported snpEff database!
        """

```

```

[Tue Mar 17 19:50:37 2026]
localrule all:
    input: results/snakefile/dirs_created, results/raw/reference.fasta, results/raw/SRR1972739.fastq, results/qc/SRR1972739_fastqc.html, results/qc/SRR1972739_fastqc.zip, results/raw/reference.dict, results/aligned/aligned.sam, results/aligned/aligned.sorted.bam, results/aligned/dedup.bam, results/aligned/dup_metrics.txt, results/variants/raw_variants.vcf, results/variants/filtered_variants.vcf, results/snpEff/data/reference_db/genes.gbk, results/snpEff/snpEff.config
    jobid: 0
    reason: Input files updated by another job: results/snpEff/snpEff.config, results/snpEff/data/reference_db/genes.gbk, results/aligned/dup_metrics.txt, results/aligned/aligned.sorted.bam, results/raw/reference.dict, results/qc/SRR1972739_fastqc.zip, results/variants/raw_variants.vcf, results/variants/filtered_variants.vcf, results/aligned/aligned.sorted.bam, results/qc/SRR1972739_fastqc.html, results/aligned/dedup.bam
    resources: tmpdir=/tmp

[Tue Mar 17 19:50:37 2026]
Finished job 0.
6 of 6 steps (100%) done
Complete log: .snakemake/log/2026-03-17T194953.247237.snakemake.log
(/home/ubuntu/BINF5506-assignment3/bioenv) ubuntu@ip-172-31-18-0:~/BINF5506-assignment3$

```

```

rule snpEff_Annotation:
    input:
        f"{SNPEFF_DIR}/snpEff.config",
        f"{VARIANT_DIR}/filtered_variants.vcf"
    output:
        f"{ANNOTATED_DIR}/annotated_variants.vcf",
        f"{SNPEFF_DIR}/snpEff.html"
    shell:
        """
        echo Annotating variants with snpEff...
        snpEff -c {SNPEFF_DIR}/snpEff.config -stats {SNPEFF_DIR}/snpEff.html reference_db {VARIANT_DIR}/filtered_variants.vcf > {ANNOTATED_DIR}/annotated_variants.vcf
        echo Annotating variants with snpEff!
        echo Pipeline completed successfully! Check the folders in {SNPEFF_DIR} for output files.
        """

```

```

Annotating variants with snpEff...
Annotating variants with snpEff!
Pipeline completed successfully! Check the folders in results/snpEff for output files.
[Tue Mar 17 22:55:41 2026]
Finished job 0.
1 of 2 steps (50%) done
Select jobs to execute...

[Tue Mar 17 22:55:41 2026]
localrule all:
    input: results/snakefile/dirs_created, results/raw/reference.fasta, results/raw/SRR1972739.fastq, results/qc/SRR1972739_fastqc.html, results/qc/SRR1972739_fastqc.zip, results/raw/reference.dict, results/aligned/aligned.sam, results/aligned/aligned.sorted.bam, results/aligned/dedup.bam, results/aligned/dup_metrics.txt, results/variants/raw_variants.vcf, results/variants/filtered_variants.vcf, results/snpEff/data/reference_db/genes.gbk, results/snpEff/snpEff.config, results/annotated/annotated_variants.vcf, results/snpEff/snpEff.html
    jobid: 0
    reason: Input files updated by another job: results/snpEff/snpEff.html, results/annotated/annotated_variants.vcf
    resources: tmpdir=/tmp

[Tue Mar 17 22:55:41 2026]
Finished job 0.
2 of 2 steps (100%) done
Complete log: .snakemake/log/2026-03-17T225538.929134.snakemake.log

```

commit fb6aed9bc97ddb8ac10988d67c2563c6a24f0574 (HEAD -> master, origin/master)

Author: BestAbrar <iloovechicken133@gmail.com>
Date: Tue Mar 17 22:58:27 2026 +0000

add "snpEff_Annotation" rule

commit 28245698c517168873d824b5e19f738e7c173f2

Author: BestAbrar <iloovechicken133@gmail.com>
Date: Tue Mar 17 19:53:43 2026 +0000

add "snpEff_database" rule

commit 2e7adcfe4756265f28814f3d619c3a7fe0433422

Author: BestAbrar <iloovechicken133@gmail.com>
Date: Tue Mar 17 19:28:54 2026 +0000

add "Variant_Calling" rule

commit 5df64ed579c409c6986ce89368293fcaadd05635

Author: BestAbrar <iloovechicken133@gmail.com>
Date: Tue Mar 17 19:25:43 2026 +0000

add "Create_BAM" rule

commit 3bcd3b6e7c709748b8d767b7f7c029d44ff6459

Author: BestAbrar <iloovechicken133@gmail.com>
Date: Tue Mar 17 19:10:47 2026 +0000

add "Create_SAM" rule

commit 7e1e7bc826ecdb8a1c9fb4f8b33fba714cbf28cc

Author: BestAbrar <iloovechicken133@gmail.com>
Date: Mon Mar 16 00:35:16 2026 +0000

add "QC_check" rule

commit 33c7e8e33fe46d7b481b565a5570389918a38f4d

Author: BestAbrar <iloovechicken133@gmail.com>
Date: Mon Mar 16 00:24:12 2026 +0000

add "download_fastq" rule

commit 286cf834ad240c5641f0d29c2debaae18b00b58a

Author: BestAbrar <iloovechicken133@gmail.com>
Date: Mon Mar 16 00:18:49 2026 +0000

add "download_fasta" rule

commit 2ad64303cbd6f3037f59757652da4cf97004f665

Author: BestAbrar <iloovechicken133@gmail.com>
Date: Mon Mar 16 00:08:41 2026 +0000

add "create_dirs" rule

commit af8e65a4b29e5303263721ac4d7b1a6131be5240

Author: BestAbrar <iloovechicken133@gmail.com>
Date: Sun Mar 15 23:19:57 2026 +0000

Upload Snakemake file

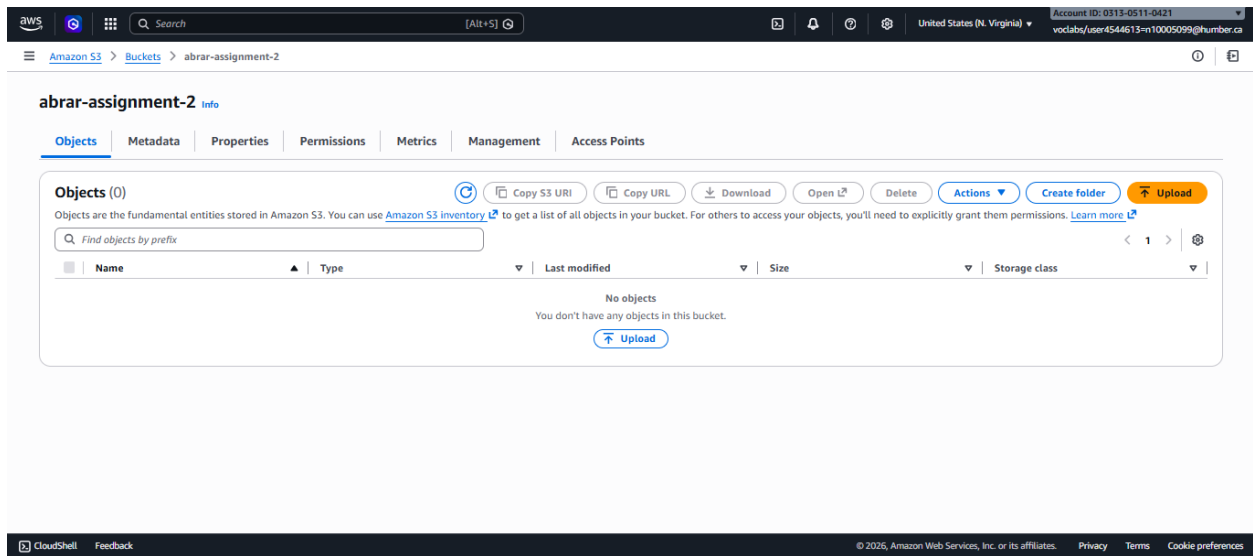
commit d73df4958d71a4021be467166f7176a10bbab751

Author: BestAbrar <iloovechicken133@gmail.com>
Date: Sun Mar 15 23:05:49 2026 +0000

Upload Snakemake file

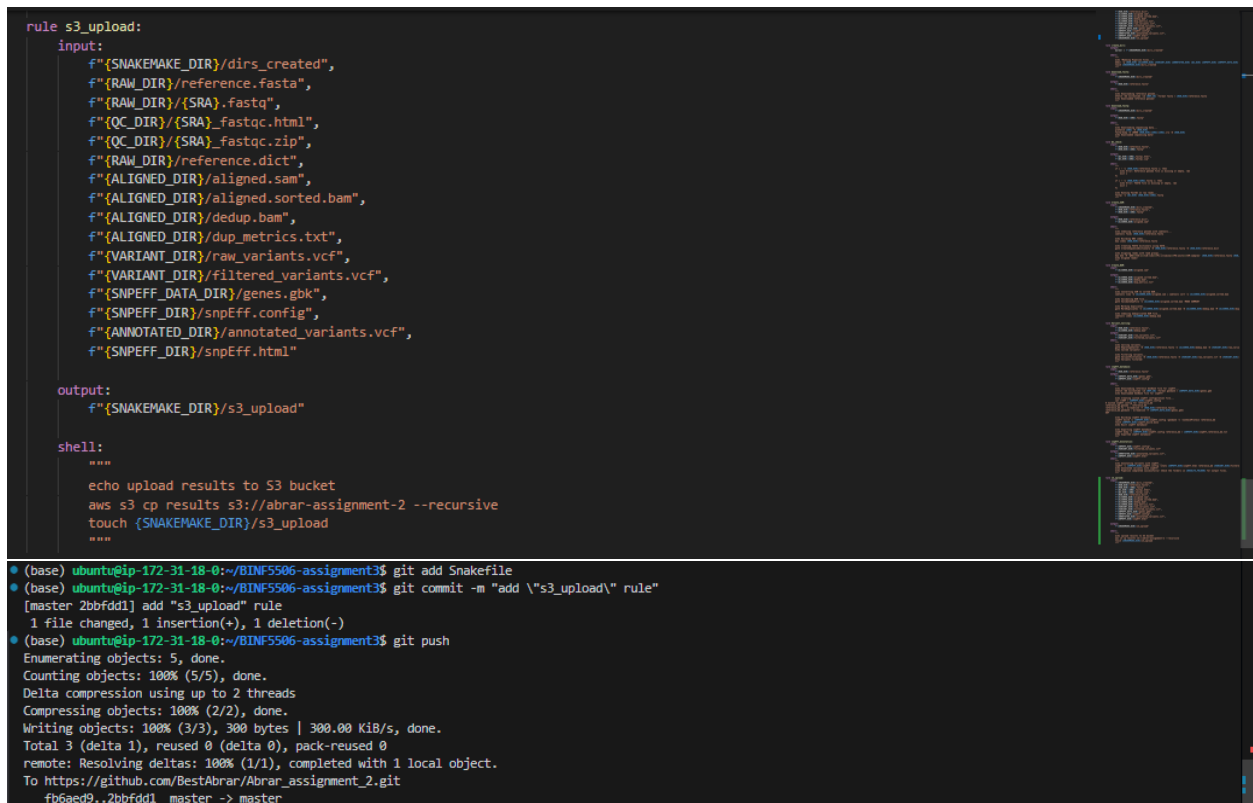
(END)

Part D:



The screenshot shows the AWS Management Console interface for the 'abrar-assignment-2' bucket. The 'Objects' tab is active, displaying a table with columns for Name, Type, Last modified, Size, and Storage class. The table is empty, with a message stating 'No objects. You don't have any objects in this bucket.' and an 'Upload' button. The console also shows the bucket's metadata, properties, permissions, metrics, management, and access points.

Part E:



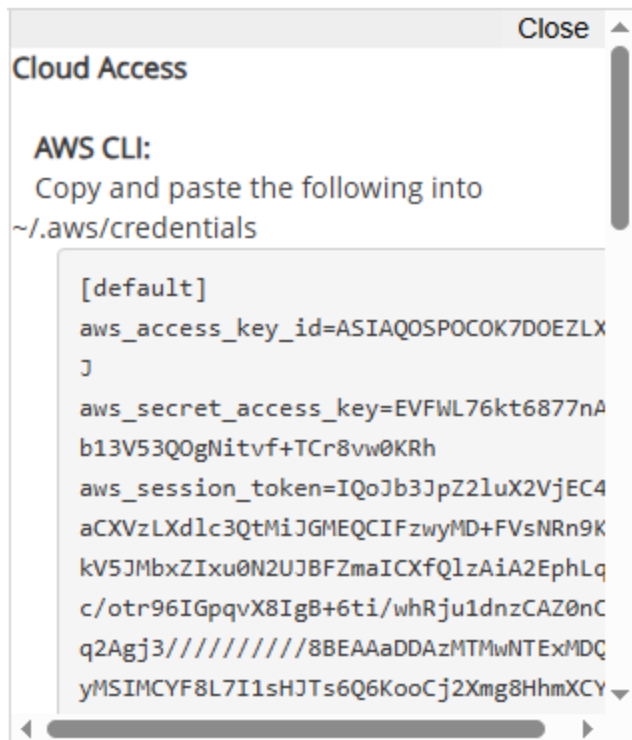
```
rule s3_upload:
    input:
        f"{SNAKEMAKE_DIR}/dirs_created",
        f"{RAW_DIR}/reference.fasta",
        f"{RAW_DIR}/{SRA}.fastq",
        f"{QC_DIR}/{SRA}_fastqc.html",
        f"{QC_DIR}/{SRA}_fastqc.zip",
        f"{RAW_DIR}/reference.dict",
        f"{ALIGNED_DIR}/aligned.sam",
        f"{ALIGNED_DIR}/aligned.sorted.bam",
        f"{ALIGNED_DIR}/dedup.bam",
        f"{ALIGNED_DIR}/dup_metrics.txt",
        f"{VARIANT_DIR}/raw_variants.vcf",
        f"{VARIANT_DIR}/filtered_variants.vcf",
        f"{SNPEFF_DATA_DIR}/genes.gbk",
        f"{SNPEFF_DIR}/snpeff.config",
        f"{ANNOTATED_DIR}/annotated_variants.vcf",
        f"{SNPEFF_DIR}/snpeff.html"

    output:
        f"{SNAKEMAKE_DIR}/s3_upload"

    shell:
        echo upload results to S3 bucket
        aws s3 cp results s3://abrar-assignment-2 --recursive
        touch {SNAKEMAKE_DIR}/s3_upload
```

The terminal output shows the successful execution of the rule and the upload of files to the S3 bucket. The output includes the following lines:

```
• (base) ubuntu@ip-172-31-18-0:~/BINFS506-assignment3$ git add Snakefile
• (base) ubuntu@ip-172-31-18-0:~/BINFS506-assignment3$ git commit -m "add \"s3_upload\" rule"
[master 2bbfdd1] add "s3_upload" rule
1 file changed, 1 insertion(+), 1 deletion(-)
• (base) ubuntu@ip-172-31-18-0:~/BINFS506-assignment3$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 300 bytes | 300.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/BestAbrar/Abra_assignment_2.git
fb6aed9..2bbfdd1 master -> master
```



```
(/home/ubuntu/BINF5506-assignment3/bioenv) ubuntu@ip-172-31-18-0:~/BINF5506-assignment3$ cat ~/.aws/credentials
[default]
aws_access_key_id=ASIAQ0SPOCOK7DOEZLXJ
aws_secret_access_key=EVFWL76kt6877nAb13V53Q0gNltvf+TCr8vw0KRh
aws_session_token=IQoJb3JpZ2luX2VjEC4aCXVzLXdlc3QtMiJGMEQCIFzwyMD+FVsNRn9Kkv5JMbXZIXu0N2UJBFZmaICXfQ1zAiA2EphLqC/otr96IGpqvX8IgB+6ti/whRju1dnzCAZ0nCq2Agj3/////////8BEAAaDDAzMTMwNTExMDQyMSIMCYF8L7I1sHJT56Q6KooCj2Xmg8HhmXCY
```

Amazon S3 > Buckets > abrar-assignment-2

abrar-assignment-2 Info

Objects (7)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	aligned/	Folder	-	-	-
☐	annotated/	Folder	-	-	-
☐	gc/	Folder	-	-	-
☐	raw/	Folder	-	-	-
☐	snakemake/	Folder	-	-	-
☐	snpEff/	Folder	-	-	-
☐	variants/	Folder	-	-	-

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